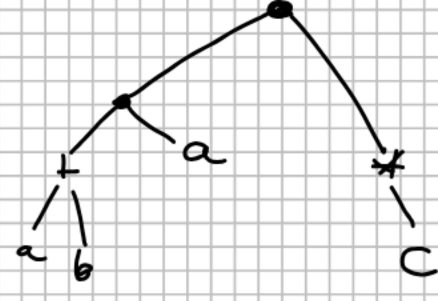
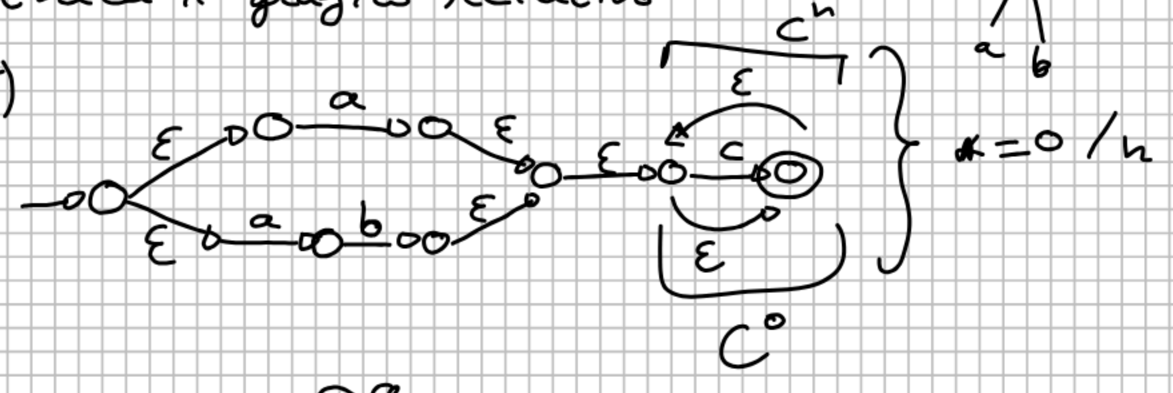
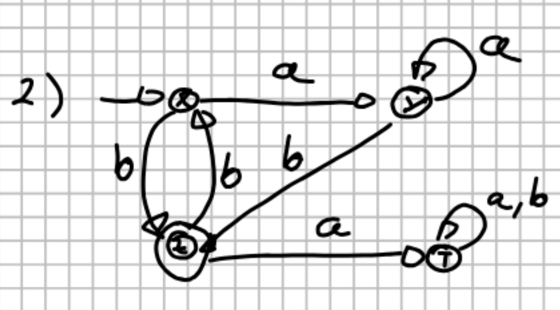
Es proposti

1) Dato il Ling $((a+ab)c^*)$ si tracci il grafico relativo





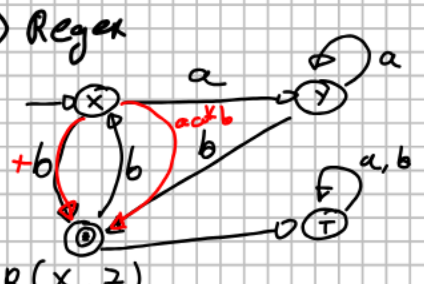


a) Si dia una gramm. lineare destra che lo genera

X	a	Y
X	b	Z
Y	a	Y
Y	b	Z
Z	a	T
Z	b	X
T	a	T
T	b	T

$X \to aY \mid bZ$
$Y \to aY \mid bZ \mid b$
$Z \to aT \mid bX \mid \varepsilon$
$T \to aT \mid bT$

2) Regex



elimina Y $\quad R(X, Z)$

$p \to q_n \to r \qquad aa^*b + b^*$

Ingressi: Y $\quad X \to a$
Uscite Y $\quad Z \to b$

3) Algo di Det per automa det

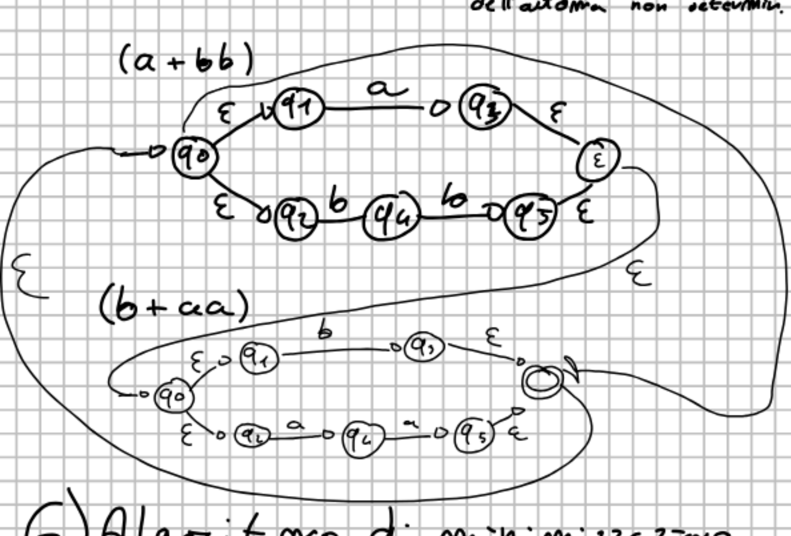
i	S_i	a	b	
0	q_0	1	2	
1	q_1,q_3,q_4	3	4	fin
2	q_0,q_2	1	5	
3	q_0,q_5	1	2	
4	q_4	6	4	fin
5	q_0,q_2,q_3,q_4	7	5	fin
6	q_5	4	$\emptyset$	
7	q_1,q_3,q_4,q_5	8	4	fin
8	q_0,q_4,q_5	7	9	fin
9	q_0,q_2,q_4	7		

4) Tabella transiz. per $\{a^{m+2n} b^m c^n \mid m,n \geq 0\}$
Automa a pila

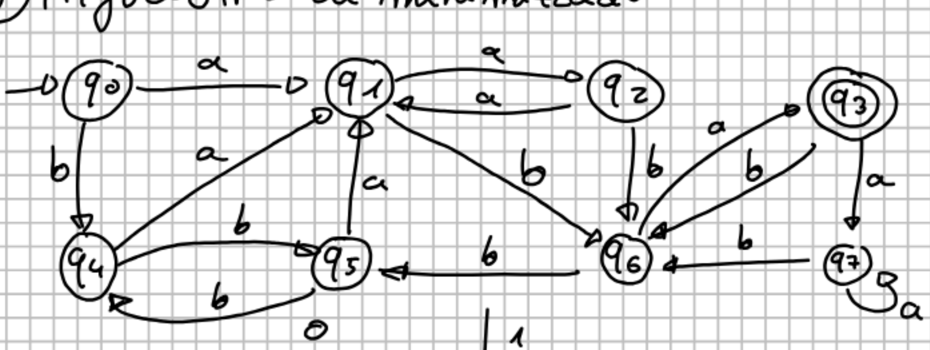
$m=2 \quad n=2 \qquad a = (b) + (2c)$

$(4+2)$
aaaaaa bb cc
aa = bb $\quad$ aaaa = cc

5) $((a+bb)(b+aa))^*$ tracciare grafico dell'automa non determin.



6) Algoritmo di minimizzazione

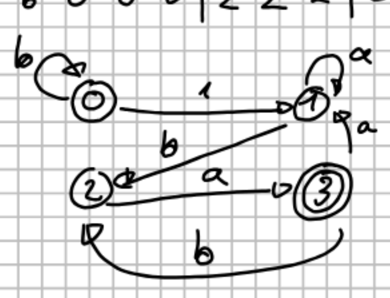


	0							1
	q_0	q_1	q_2	q_4	q_5	q_6	q_7	q_3
a	0	0	0	0	0	1	0	0
b	0	0	0	0	0	0	0	0

	0						1	2
	q_0	q_1	q_2	q_4	q_5	q_7	q_6	q_3
a	0	0	0	0	0	0	2	0
b	0	1	1	0	0	1	0	1

	0			1			2	3
	q_0	q_4	q_5	q_1	q_2	q_7	q_6	q_3
a	1	1	1	1	1	1	3	1
b	0	0	0	2	2	2	0	2

algo finito non posso più spezzare in blocchi



7) Grammatica

$S \to aXb \qquad X \to YZ \mid bSa \mid \varepsilon$
$Y \to a \mid \varepsilon \qquad Z \to YZ$

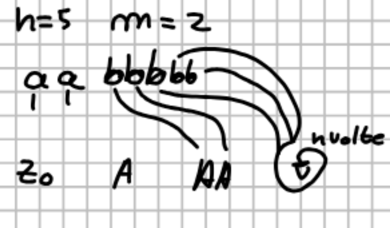
1) Annullabili $\{X \text{ e } Y\}$
$S \to aXb \mid ab \qquad X \to YZ \mid Z \mid bSa$
$Y \to a \qquad Z \to YZ$

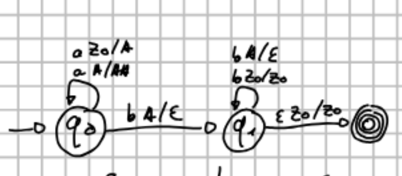
2) Unarie se $A \to a$ a var
$X \to Z$ ciò ci fa avere $X \to YZ$ ma già presente in X quindi è inutile. Mi basta quindi togliere Z da X
$S \to aXb \mid ab \qquad X \to YZ \mid bSa$
$Y \to a \qquad Z \to YZ$

3) Determinare le produttive e accessibili
S → Produttiva $\quad$ X → produttiva ma non $X \to YZ$
Y → produttiva $\quad$ Z → non produttiva
tolgo Z perché genera solo parole nonfinite
Di conseguenza Y diventa inaccessibile e tolgo pure lei. La grammatica finale sarà
$S \to aXb \mid ab \qquad X \to bSa$

8) Automa a pila di

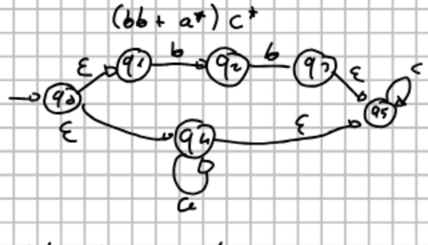
$\{a^m b^n \mid n > m > 0\}$
$n=5 \quad m=2$



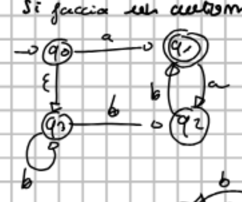


	a	b	ε
$q_0\,Z_0$	$q_0\,A$	–	–
$q_0\,A$	$q_0\,AA$	$q_1\,\varepsilon$	–
$q_1\,Z_0$	–	$q_1\,Z_0$	$f\,Z_0$
$q_1\,A$	–	$q_1\,\varepsilon$	–

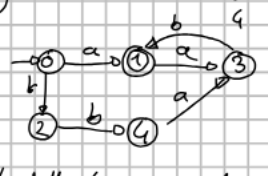
9) grafo di un automa che riconosca
$(bb+a^*)c^*$



10) Utilizzando la costruzione dei sottoinsiemi si faccia un automa minimo equivalente



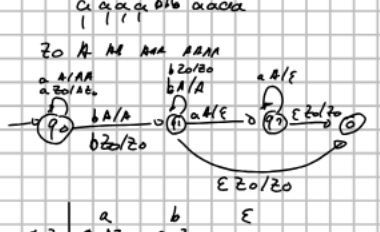
i	S_i	a	b	
0	$q_0\,q_3$	1	2	
1	q_1	3	$\emptyset$	✓
2	$q_3\,q_2$	$\emptyset$	4	
3	q_2	$\emptyset$	1	
4	$q_3\,q_2\,q_1$	3	4	✓



11) tabella transizioni automa a pila che riconosca

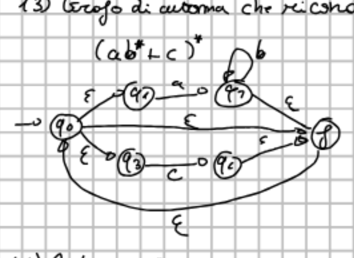
$\{a^m b^n a^m \mid m, n \geq 0\}$
$m=4 \quad n=3$
aaaa bbb aaaa

Z_0 A AA AAA AAAA

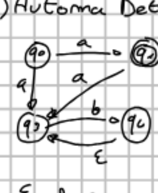


	a	b	ε
$q_0\,Z_0$	$q_0\,AZ_0$	$q_1\,Z_0$	–
$q_0\,A$	$q_0\,AA$	$q_1\,A$	–
$q_1\,Z_0$	–	$q_1\,Z_0/Z_0$	$q_f\,Z_0$
$q_1\,A$	$q_2\,\varepsilon$	$q_1\,A/A$	–
$q_2\,Z_0$	–	–	$q_f\,Z_0$
$q_2\,A$	$q_2\,A$	–	–

13) Grafo di automa che riconosca
$(ab^*+c)^*$



14) Automa Deterministico relativo a

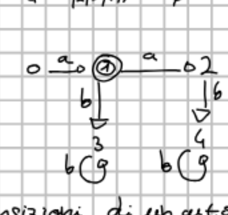


Usando la costruzione di sottoinsiemi

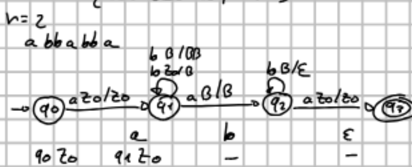
ε chiusure

i	$C_\varepsilon(q_i)$
0	q_0
1	q_1
2	$q_2\,q_5$
3	q_3
4	$q_3\,q_4$
5	q_5

i	S_i	a	b	
0	q_0	1	$\emptyset$	
1	$q_1\,q_3$	2	3	X
2	$q_2\,q_3\,q_5$	$\emptyset$	4	
3	$q_3\,q_4$	$\emptyset$	3	
4	$q_2\,q_3\,q_4\,q_5$	$\emptyset$	4	



15) Tabella transizioni di un automa a pila per
$\{a b^n a b^n a \mid n > 0\}$
$n=2$
abbabba



	a	b	ε
$q_0\,Z_0$	$q_1\,Z_0$	–	–
$q_1\,Z_0$	–	$q_1\,B$	–
$q_1\,B$	$q_2\,B$	$q_1\,BB$	–
$q_2\,Z_0$	$f\,Z_0$	–	–
$q_2\,B$	–	$q_2\,\varepsilon$	–

12) Grammatica in Chomsky a partire da
$S \to XY \qquad X \to aSb \mid Y \qquad Y \to X \mid \varepsilon$

1) Annullabili
$Y \to \varepsilon \qquad X \to Y \to \varepsilon \qquad S \to XY \to \varepsilon\varepsilon \mid X \mid Y$
$[S \to XY \mid X \mid Y \mid \varepsilon \qquad X \to aSb \mid ab \mid Y \qquad Y \to X$

2) Unarie
$S \to X \quad S \to Y \quad X \to Y \quad Y \to X$
$S \to XY \mid \underbrace{aSb \mid ab}_{X} \mid \underbrace{\cancel{aSb \mid ab}}_{Y} \mid \varepsilon \qquad X \to aSb \mid ab$
$S \to XY \mid aSb \mid ab \mid \varepsilon \qquad X \to aSb \mid ab \qquad Y \to aSb \mid ab$

3) tutte produttive e accessibili

Per portarlo in Chomsky:
$A \to a \quad B \to b \quad T \to AS$ (oppure $T \to SB$)
$S \to XY \mid TB \mid AB \qquad X \to TB \mid AB \qquad Y \to TB \mid AB$

16) Portare in Chomsky
$S \to Xa \qquad X \to YY \mid Xb \qquad Y \to a \mid \varepsilon$

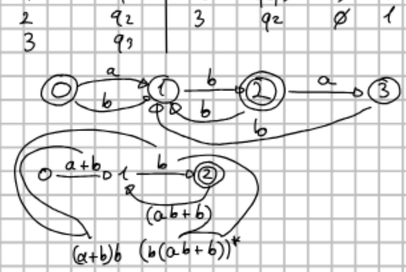
1) Annullabili
Y annullabile, per propagazione $X \to YY \to \varepsilon$
$S \to Xa$ avrà sempre un terminale quindi non è annullabile, ma dobbiamo tenere in considerazione che $X \to YY \to \varepsilon\varepsilon \Rightarrow S \to \varepsilon a \Rightarrow S \to a$

22) Algo di determinizzazione

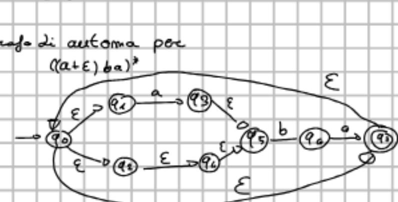
Chiusure

ε	$C_\varepsilon(q_i)$
0	$q_0\,q_2$
1	q_1
2	q_2
3	q_3

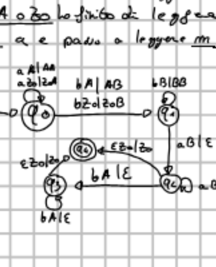
i	S_i	a	b	F
0	$q_0\,q_2$	1	1	
1	q_1	$\emptyset$	$q_2\,q_3$	
2	$q_2\,q_3$	3	1	V
3	q_2	$\emptyset$	1	



21) Grafo di automa per
$((a+\varepsilon)ba)^*$



20) automa a pila per
$\{a^m b^n a^n b^m \mid n > 0\}$
$m=2 \quad b=3$
aa bbb aaa bb
Z_0, A, AA
Testa Z_0 o A
$\Rightarrow$ BBB $\Rightarrow$ se testa = B
toglie B (ε)
Altrimenti se leggo
A o Z_0 ho finito di leggere
n a e passo a leggere m b



	a	b	ε
$q_0\,Z_0$	q_0,Z_0A	q_1,Z_0B	–
$q_0\,A$	q_0,AA	q_1,AB	–
$q_1\,B$	q_2,ε	q_1,BB	–
$q_2\,Z_0$	–	–	q_1,Z_0
$q_2\,A$	–	q_3,ε	–
$q_2\,B$	q_2,ε	–	–
$q_3\,Z_0$	–	–	q_4,Z_0
$q_3\,A$	–	q_3,ε	–
$q_4\,f$			